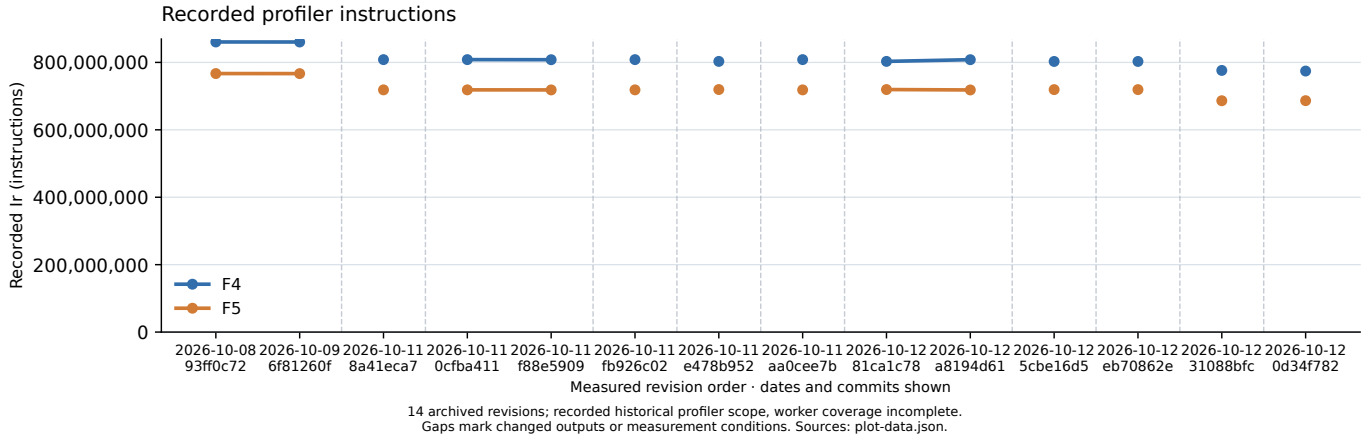
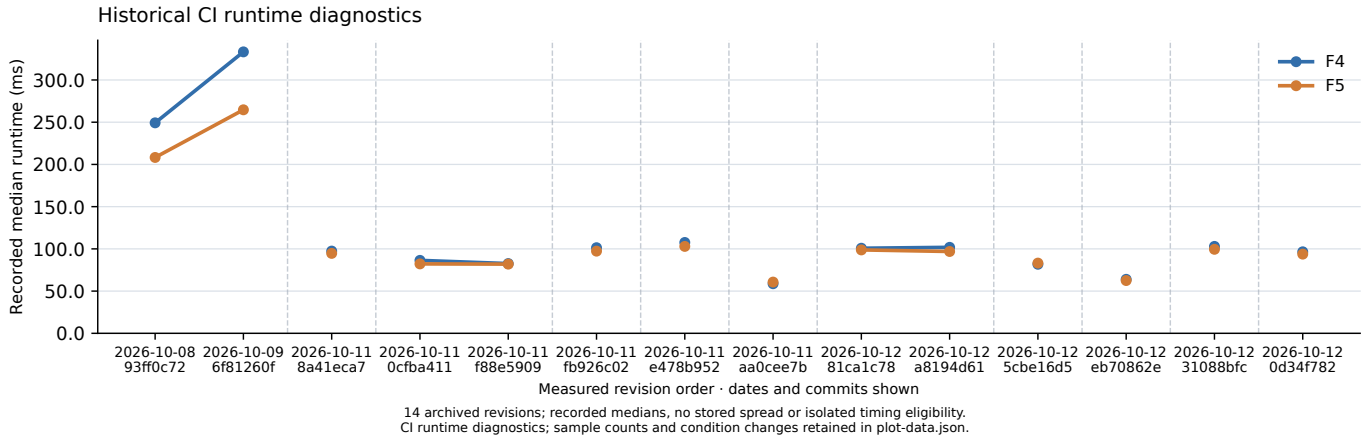
# Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.




| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 860,633,889 | 766,940,469 | 249.241 | 208.250 |
| 2026-10-09 / 6f81260f | 860,555,567 | 766,776,087 | 333.279 | 264.687 |
| 2026-10-11 / 8a41eca7 | 808,263,705 | 718,576,235 | 97.380 | 94.658 |
| 2026-10-11 / 0cfba411 | 808,263,705 | 718,578,523 | 86.390 | 82.214 |
| 2026-10-11 / f88e5909 | 807,989,935 | 718,391,055 | 82.597 | 81.948 |
| 2026-10-11 / fb926c02 | 808,263,705 | 718,576,189 | 101.379 | 97.323 |
| 2026-10-11 / e478b952 | 803,017,657 | 719,516,063 | 107.585 | 102.927 |
| 2026-10-11 / aa0cee7b | 808,281,869 | 718,492,761 | 58.723 | 60.673 |
| 2026-10-12 / 81ca1c78 | 803,015,877 | 719,509,003 | 100.762 | 98.775 |
| 2026-10-12 / a8194d61 | 808,072,539 | 718,282,030 | 101.827 | 96.904 |
| 2026-10-12 / 5cbe16d5 | 802,824,065 | 719,316,911 | 81.784 | 83.144 |
| 2026-10-12 / eb70862e | 802,824,065 | 719,316,876 | 63.944 | 62.505 |
| 2026-10-12 / 31088bfc | 776,242,363 | 686,315,894 | 102.929 | 99.489 |
| 2026-10-12 / 0d34f782 | 774,379,185 | 686,521,131 | 96.521 | 93.790 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 14 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.